



Virus-Induced Gene Silencing in Poaceae Using a *Foxtail Mosaic Virus* Vector

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Abstract

Virus-induced gene silencing (VIGS) is a powerful tool for rapidly knocking down the expression of plant genes to elucidate functional genomics. We have established a VIGS vector for monocot plants derived from *Foxtail mosaic virus* (FoMV), a positive-sense single-stranded RNA virus. For silencing a targeted gene, plant gene fragment was inserted into the vector between open reading frame 4 (ORF4) and ORF5 under the control of a duplicated coat protein promoter. Plants of different monocot species were infected by mechanical inoculation with sap from FoMV derivative-infected *Chenopodium quinoa* leaves. Gene silencing was typically observed within 2–3 weeks after inoculation. In this chapter, we describe the detailed protocol for silencing a target gene in various Poaceae plants by using FoMV-based vectors.

Key words Posttranscriptional gene silencing (PTGS), Virus-induced gene silencing (VIGS), *Foxtail mosaic virus* (FoMV), FoMV-based VIGS vector, Monocot, Poaceae

1 Introduction

With the advancement of whole-genome sequencing, a vast amount of sequence information has become available. Virus-induced gene silencing (VIGS) provides a means to unravel the functions of candidate genes identified in large genomic or transcriptomic screens [1]. Upon virus infection, plants defend themselves by activating antiviral RNA silencing pathways. These pathways involve the ribonucleases dicer-like protein (DCL) and Argonaute protein. DCL cleaves double-stranded RNAs (dsRNAs) such as self-folded viral genomes or dsRNAs of replication intermediates into small interfering RNA (siRNA) fragments of about 21–24 nt [2]. Subsequently, one strand of the siRNA is loaded into Argonaute to form an RNA-induced silencing complex that targets complementary viral RNAs for inactivation by slicing or translational inhibition [3, 4]. Additionally, the plant's RNA-dependent RNA polymerase 6 uses siRNAs as primers to synthesize new

dsRNAs for amplifying gene-silencing signals, which can spread from the site of infection throughout the entire plant to trigger systemic gene silencing [5, 6].

In VIGS, the plant is infected with an engineered virus that carries a fragment of a plant gene. The recombinant virus replicates and moves throughout the plant and triggers gene silencing to cleave the corresponding mRNA transcripts. VIGS has become a widely used technique for analyzing plant gene functions because it is easy to operate and rapid, with high specificity and high efficiency [7].

Eight plant viruses have been used for developing viral vectors for VIGS in Poaceae plants: *Barley stripe mosaic virus* in barley (*Hordeum vulgare*), *Brachypodium distachyon*, and wheat (*Triticum aestivum*) [8–11]; *Brome mosaic virus* in rice (*Oryza sativa*), barley, and maize (*Zea mays*) [12]; *Rice tungro bacilliform virus* in rice [13, 14]; *Bamboo mosaic virus* and its associated satellite RNA in *B. distachyon* [15]; *Cucumber mosaic virus* in maize [16]; *Foxtail mosaic virus* (FoMV) in maize, barley, wheat, and foxtail millet (*Setaria italica*) [17–19]; *Tobacco rattle virus* in wheat and maize [20]; and, most recently, *Chinese wheat mosaic virus* in wheat [21]. Among them, FoMV has the following advantages as an applicable VIGS vector for studying gene function in Poaceae: (1) it has a broad host range in Poaceae, including 56 species [22, 23], (2) it induces only mild symptoms in native or adaptive hosts [17, 18], and (3) preparing the inoculum is very easy.

FoMV, a member of the genus *Potexvirus* in the family *Alphaflexiviridae*, has a single-stranded, positive-sense RNA genome of approximately 6200 nt [excluding the 3' poly(A) tail]. The five open reading frames (ORF1 to 5) of the FoMV genome encode ORF1 protein (152 kDa), triple-gene-block protein 1 (TGBp1, 26 kDa), TGBp2 (11 kDa), TGBp3 (6 kDa), and capsid protein (CP, 24 kDa), respectively. Unlike other potexviruses, FoMV has an additional ORF5A, which initiates 144 nt upstream of ORF5 and causes an N-terminal CP extension with dispensable function in FoMV infection (Fig. 1a) [24]. ORF1 encodes a viral replicase that is expressed from genomic RNA (gRNA), whereas TGBp1-3 and CP are expressed from subgenomic RNA1 (sgRNA1) and sgRNA2, respectively. The synthesis of sgRNAs is individually driven by subgenomic promoter 1 (SGP1) and cpSGP [25].

Two FoMV vectors for VIGS had been applied in monocot plants. In the first one, FoMV infectious cDNA was cloned under the control of the *Cauliflower mosaic virus* (CaMV) 35S promoter and the nopaline synthase terminator. The vector was designed by engineering the cloning sites after the stop codon of ORF5. This vector successfully silenced endogenous genes in maize by delivering the plasmid of recombinant FoMV by bombardment [18]. The second cloning strategy involved duplicating cpSGP between ORF4 and ORF5 of FoMV vector for expressing homologous

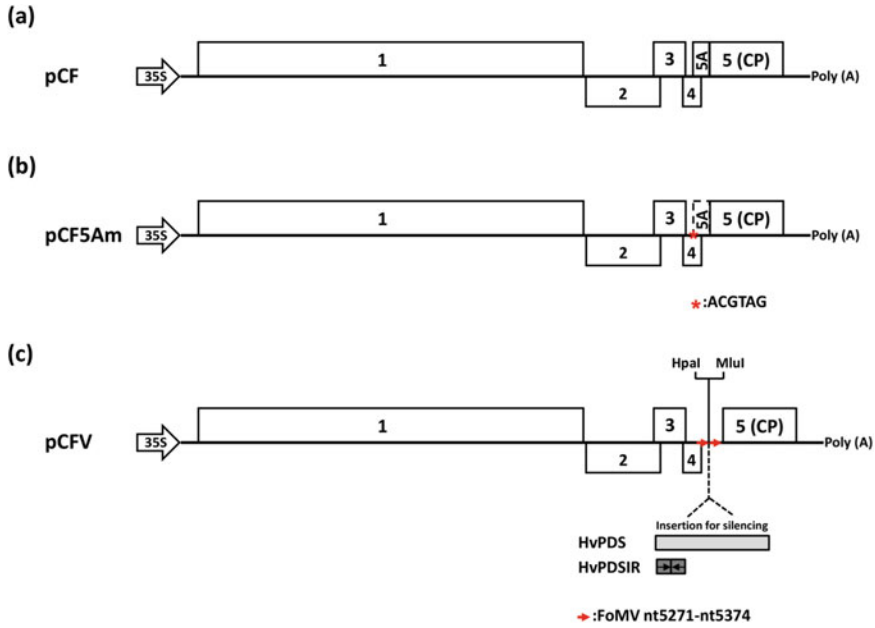


Fig. 1 Schematic representation of construction of the FoMV cDNA infectious clone and derivative mutants. (a, b) Genome organization of FoMV (pCF) and FoMV-derived mutant (pCF5Am). Rectangles represent open reading frames (ORFs), ORF1-ORF5, encoded by FoMV genomic RNA. The FoMV genome is cloned under control of the CaMV 35S promoter (35S). Red star (*) in frame B: the start codon and second codon of ORF5A were altered to ACGTAG for disrupting translation of 5A protein. (c) Schematic diagram of the constructed FoMV silencing vector, pCFV. Plasmid of pCFV was derived from pCF5Am and duplicated the promoter region of CP (red arrowhead). Gene fragments for silencing are inserted between the cut sites of HpaI and MluI. The location of the CP promoter in FoMV genome is indicated below the diagram

fragments of host genes. Endogenous genes in barley, wheat, and foxtail millet were efficiently silenced after sap inoculation of recombinant FoMV in agroinfiltrated *Nicotiana benthamiana* leaves [17].

Here, we present the use of a new FoMV vector pCFV designed for VIGS. The cDNA of the FoMV genome (AY121833.1) was cloned under the control of the CaMV 35S promoter to generate pCF (Fig. 1a). To develop an FoMV VIGS vector, we employed the strategy of cpSGP duplication between ORF4 and ORF5. Because duplicated cpSGP and insertion of foreign fragments would interfere with the translation codon of protein 5A, we stopped the expression of protein 5A by substitution of the initiation and second codons of ORF5A with ACGTAG and named as pCF5Am (Fig. 1b). The final vector pCFV is a pCF5Am derivative that contains a putative FoMV cpSGP (nt5271-nt5374 of AY121833.1) and cloning sites (HapI and MluI) inserted between ORF4 and ORF5 (Fig. 1c) [26]. In this chapter, we present a detailed protocol for constructing the target sequence into pCFV and silencing a target gene in various Poaceae plants by using an

FoMV-based vector. It may be an efficient strategy for high-throughput functional genomics in important cash crops.

2 Materials

2.1 Plant Materials and Plant Growth

1. *Brachypodium distachyon* seeds: lines Bd3-1, Bd21-1, and Bd21-3.
2. *Chenopodium quinoa* seeds.
3. *Hordeum vulgare* seeds.
4. *Oryza sativa* L. seeds: Taikeng No. 9, Taitung Glutinous No. 31, Taoyuan No. 3, and Tatsuen No. 2.
5. *Setaria viridis* seeds.
6. *Sorghum bicolor* seeds: Taichung No. 3.
7. *Sorghum sudanense* seeds.
8. *Triticum aestivum* seeds.
9. Soil mix and pots.
10. Greenhouse at 25 °C under a 16-h/8-h light/dark photoperiod.

2.2 Creating Silencing Constructs

1. *Escherichia coli* strain DH5α competent cells.
2. pCFV-based expression vector [26]: an infectious cDNA clone of FoMV (AY121833.1) driven by the 35S promoter of *Cauliflower mosaic virus* (CaMV) and containing a duplicated cpSGP and cloning sites (HapI and MluI) inserted between ORF4 and ORF5 (Fig. 1c).
3. Restriction enzymes and buffers: *HpaI* and *MluI*.
4. 1% Agarose gel containing 0.5 mg/mL of ethidium bromide, 0.5× TBE buffer (44.5 mM Tris-HCl pH 8.0, 44.5 mM boric acid, 1 mM EDTA), and electrophoresis equipment.
5. Gel Extraction Kit and PCR Purification Kit.
6. cDNAs of Poaceae plants.
7. Oligonucleotide primers for amplifying target sequences.
8. *Taq* DNA polymerase and buffer.
9. T4 DNA ligase and buffer.
10. Luria Broth (LB) liquid media and agar plates.
11. 50 mg/mL Ampicillin stock solution.
12. High-volume plasmid purification kit (e.g., NucleoBond Xtra Maxiprep kit, Macherey-Nagel).
13. Thermal cycler.
14. 37 and 16 °C water bath.

15. Fo5137F primer 5'-GCATGCACGAATCACAC-3' and Fo5655R primer 5'-GCTAGCGCCAGACTTTTC-3'.

2.3 Plant Inoculation with FoMV Plasmids

1. pCFV Empty vector plasmid (Fig. 1c).
2. pCFHvPDS or pCFHvPDSIR (*H. vulgare phytoene desaturase* target sequence) as a positive control for VIGS [26] (Fig. 1c, see Note 1).
3. Recombinant pCFV plasmids with target sequence insertion.
4. Carborundum.
5. Cotton swabs.
6. 0.01 M sodium phosphate, pH 7.0.
7. Pestle and mortar.
8. Celite 545.

2.4 Validation of Silencing

1. Total RNA extraction reagent (e.g., TriPure Isolation Reagent, Roche Life Science).
2. Reverse transcription kit including oligo dT(18) primer.
3. qPDSF Primer 5'-AATCCTCCTGAAAGGCTATG-3' and qPDSR primer 5'-TCTCCAGTTATTTGAGTCCC-3'.
4. qActinF Primer 5'-GAAGATCCTCACCGAGAGAGGTTA-3' and qActinR primer 5'-GTCACACTTCATGATGGAGTTGTA-3'.
5. SYBR Green Master Mix.
6. qPCR Thermal cycler.

3 Methods

3.1 Construction of pCFV Containing the Target Gene for Silencing

3.1.1 Preparation of Target Gene Fragment

1. Select a 300–500 base pair (bp) region of the target gene for silencing (see Note 2).
2. Design oligonucleotide primers to amplify the target region for silencing. The primers should be designed to include the site *HpaI* and/or *MluI* at the 5' end of the forward and reverse primer, respectively, for cloning into pCFV.
3. Amplify the target region by PCR using the oligonucleotide primers, Taq DNA polymerase, and cDNA of *Poaceae* plant as template in 25 µL volume.
4. Electrophorese 3 µL of the PCR product on 1% agarose gel to confirm amplification of the correct-sized product. Use the rest of the sample to purify the amplified PCR fragment by a PCR cleanup kit.
5. Digest 1 µg of the purified PCR fragment with *HpaI* and/or *MluI* according to the design.

6. Electrophorese the digested PCR product on 1% agarose gel, and purify the band by using a gel extraction kit.

3.1.2 Preparation of pCFV Vector for Cloning

1. Digest 2 µg of pCFV plasmid with *HpaI* and/or *MluI* according to the design.
2. Electrophorese the digested product on 1% agarose gel, and purify the band by using a gel extraction kit.

3.1.3 Cloning of pCFV Containing the Target Gene

1. Ligate the digested PCR product (insert fragment) into digested pCFV (the vector) using a 1:3 molar ratio of vector/insert DNA in 10 µL volume. Incubate reaction at room temperature for 1 h.
2. Transform the ligation product into competent *E. coli* DH5α by heat shock.
3. Spread the cells onto LB agar plates supplemented with 50 µg/mL ampicillin, and incubate overnight at 37 °C.
4. Confirm the presence of pCFV carrying the target gene sequence by colony PCR with the primers Fo5137F and Fo5655R on individual transformants.
5. Positive clones are grown in 2 mL LB liquid containing 50 µg/mL ampicillin overnight at 37 °C.
6. Extract plasmid DNAs of the confirmed silencing constructs by using a miniprep procedure and digest with *HpaI* and/or *MluI* to determine that the target fragment for silencing is present.
7. Sequence the positive clones with Fo5137F sequencing primer (see **Note 3**).
8. Prepare the silencing construct in large-scale cultures by using a high-volume plasmid purification kit to minimum concentration of 2 µg plasmid per mL culture.

3.2 Plant Inoculation with FoMV Plasmids

3.2.1 Mechanical Inoculation of FoMV Plasmid into *C. quinoa*

1. Sow *C. quinoa* seeds in pots with soil mix. Place the pots in a greenhouse at 25 °C under a 16-h/8-h light/dark photoperiod. Plants are ready for inoculation at five-leaf stage (approximately 30 days old under our condition).
2. Dust every *C. quinoa* leaf with carborundum.
3. Rub 10 µL of each 200 ng/µL pCFV (negative control), pCFHvPDS or pCFHvPDSIR (positive controls), and pCFV carrying the target gene onto *C. quinoa* leaves by using a cotton swab (see **Note 4**). After 5 min the inoculated leaf was washed with sterile water to remove carborundum.
4. Symptoms of infection (local lesions) are typically observed between 7 and 14 days postinoculation on inoculated *C. quinoa* leaves (Fig. 2).

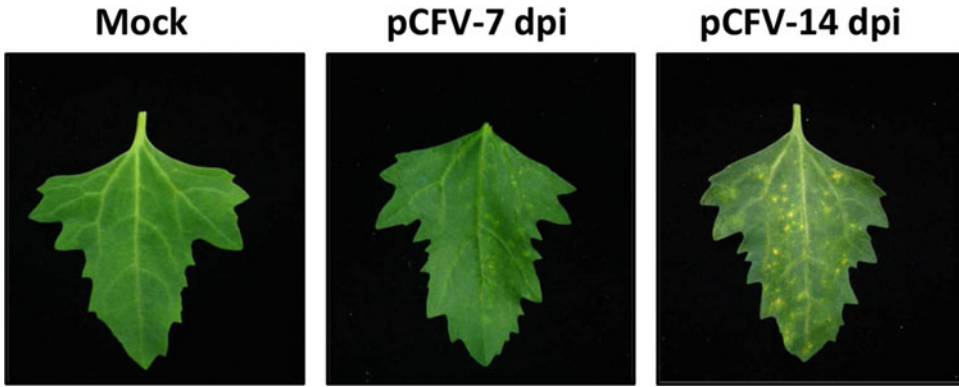


Fig. 2 Local lesions of *Chenopodium quinoa* inoculated with water (Mock) or FoMV infectious clone, pCFV. The photographs were taken at 7 days postinoculation (dpi) and 14 dpi

5. Harvest the *C. quinoa* leaves showing FoMV infection as inoculum.

3.2.2 Mechanical Inoculation of *C. quinoa* Sap into Poaceae Plants

1. Germinate seeds of Poaceae plants in 3-in. diameter pots containing the soil (*see Note 5*). Place the pots in a greenhouse at 25 °C under a 16-h/8-h light/dark photoperiod. Seedlings at the two-leaf stage are ready for inoculation.
2. Prepare VIGS inoculum by grinding infected leaf tissue of *C. quinoa* in five volumes of 0.01 M sodium phosphate buffer (pH 7.0) by using a pestle and mortar. Add 10 mg/mL Celite to the mixtures.
3. Inoculate two leaves of seedlings by rubbing with the inoculum mixture between the thumb and index finger. Care is taken to not use too much pressure that would damage the leaf.
4. Place the rub-inoculated plants in a greenhouse at 25 °C under a 16-h/8-h light/dark photoperiod.

3.2.3 Validation of Silencing

1. Isolate total RNA from the leaf tissues showing the desired phenotype by using RNA extraction reagent, as per the manufacturer's instructions. If there is no obvious phenotype, harvest the fourth leaf of infected plants after 14 or 21 days of VIGS for RNA isolation. For *PDS*-silenced plants (positive controls), the leaves should show obvious photo-bleaching in various Poaceae plants after 3 weeks (Fig. 3a).
2. Synthesize first-strand cDNA by using the oligo dT(18) primer and a reverse transcriptase according to the manufacturer's protocol.
3. Use real-time quantitative PCR (qPCR) to test for the targeted gene silencing with designed primers (*see Note 6*). Prepare qPCR reaction by mixing 6.2 µL PCR-grade water, 0.4 µL forward primer (10 mM stock), 0.4 µL reverse primer

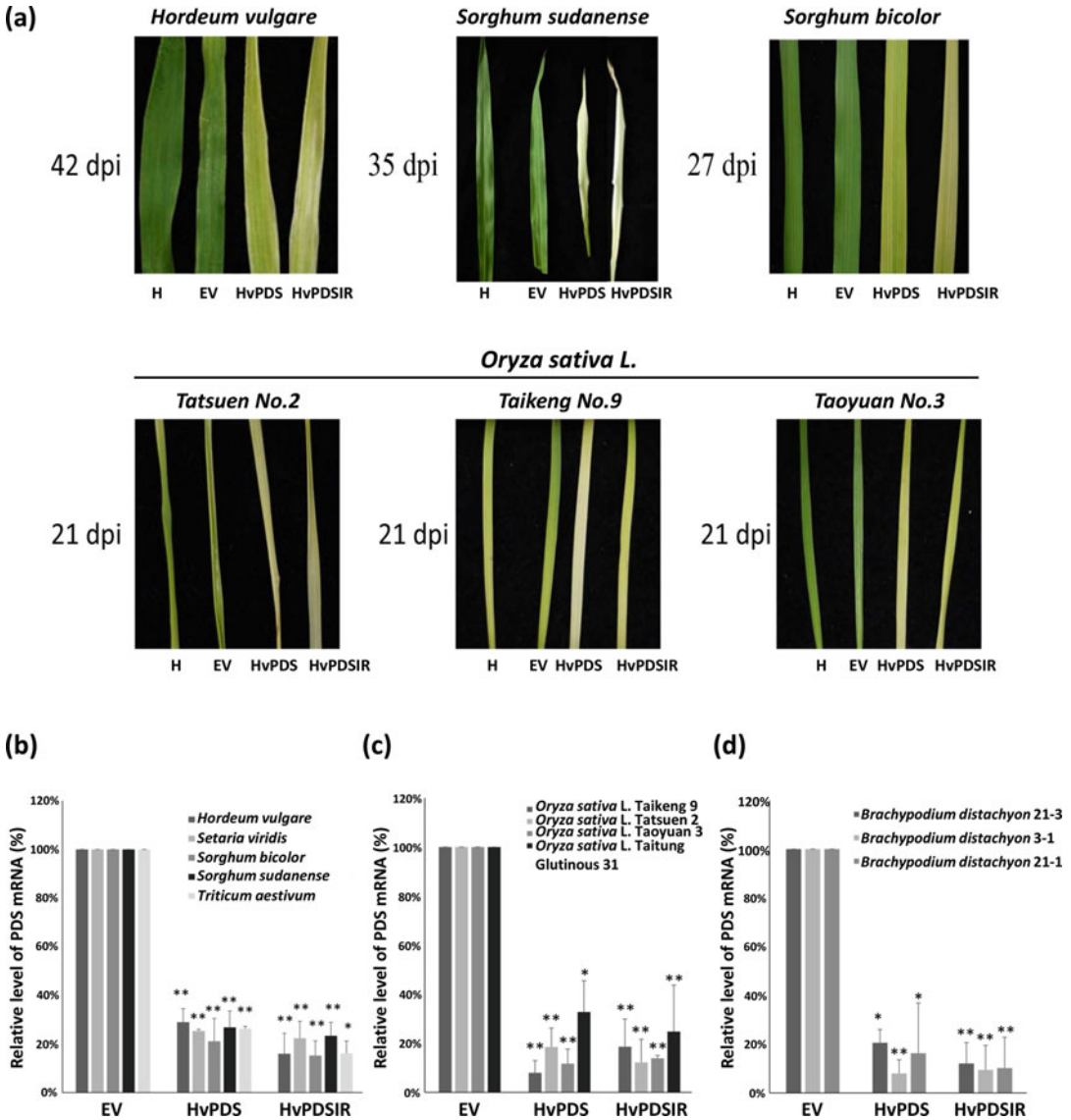


Fig. 3 PDS silencing efficiency of FoMV-VIGS vector in various Poaceae plants. (a) Phenotypes in PDS-silenced plants of barley (*Hordeum vulgare*), sudangrass (*Sorghum sudanense*), sorghum (*Sorghum bicolor*), and rice (*Oryza sativa L.*, Taikeng No. 9, Tatsuen No. 2, and Taoyuan No. 3) at 42, 35, 27, and 21 dpi, respectively. (b–d) Relative accumulation of PDS mRNA in control (EV) and PDS knockdown (HvPDS or HvPDSIR) plants measured by real-time RT-qPCR. Data are mean $\pm$ SD from three independent experiments relative to the control (EV). The mRNA level of PDS was normalized to that of *Actin*. * and ** indicate statistically significant differences by Student's *t*-test (* P < 0.05; ** P < 0.01)

(10 mM stock), 3 μ L cDNA, and 10 μ L SYBR green Master Mix in each PCR tubes. Carry out PCR according to the following program: 1 cycle of 3 min at 95 $^{\circ}$ C (enzyme activation); 40 cycles of 3 s at 95 $^{\circ}$ C (denaturation) and 20 s at 60 $^{\circ}$ C

(annealing/extension). For *PDS*-silenced plants, *PDS* is quantified by real-time RT-qPCR with primers (qPDSF and qPDSR) that are conserved in different *Poaceae* plants and specific to *PDS*.

4. Analyze the data by the $2^{-\Delta\Delta CT}$ method [27].
5. Normalize the gene of interest to the endogenous control gene, *actin*. The *actin* gene is amplified with primers (qActinF and qActinR) that can be used for various *Poaceae* plants. For *PDS*-silenced plants, *PDS* expression in FoMVPDS- and FoMVPDSIR-infected plants was decreased to 8–33% and 9–15% of that of control plants (infected with FoMV vector, pCFV) at 14 days postinoculation (Fig. 3b–d).

4 Notes

1. To induce *PDS* silencing in different plant species by using one construct, an HvPDS fragment located at nt1402-nt1843 of *H. vulgare phytoene desaturase* mRNA partial cds (GenBank: AY062039.1) was selected for cloning into the pCFV vector. This fragment shares 95%, 92%, and 90% sequence similarity with that of *T. aestivum* (FJ517553), *B. distachyon* (FJ913272), and *Oryza sativa* L. (AF049356), respectively. The high similarity (more than 88%) has the opportunity to trigger gene silencing in different plants [8]. For the HvPDSIR, an inverted repeat fragment of nt1090-nt1149 of HvPDS cDNA, shares 95%, 88%, and 89% sequence similarity with that of *TaPDS*, *BdPDS*, and *OsPDS*, respectively, was cloned into pCFV.
2. Because the region and length of the insert fragments used for silencing vary depending on the gene, optimization is required. A fragment of >300 bp is recommended for use for silencing, and if the fragment is <300 bp, the silencing efficiency would be reduced [28]. Multiple sequence alignments can be used to determine the optimal sequence for silencing. To avoid non-specific targeting, a gene-specific region should be selected for primer design. If the complete sequence of the gene is available, its untranslated region should be used for silencing. To silence all or several members of a gene family, a conserved sequence with more than 88% similarity should be selected.
3. The Fo5137F sequence primer is located 237 bases upstream of the cloning site in the pCFV vector.
4. We recommend always infection with pCFHvPDS or pCFHvPDSIR as the positive control. Suppression of *PDS* expression will result in plants exhibiting a photo-bleached phenotype that is easily observed by the naked eye and is

quite different from viral symptoms. The inverted repeat (IR) structure produced by the virus always induces better efficiency of VIGS; we recommend pCFHvPDSIR as a positive control [29].

5. Among the tested Poaceae plants in this report, only the seeds of *O. sativa* and *S. bicolor* need pretreatment to break dormancy and improve seed growth. Soak *O. sativa* seeds and *S. bicolor* seeds in water at room temperature for 3 days. Cover seeds with water at a depth of approximately 15 mm, and expose them to 28 °C for 2 days prior to sowing.
6. The primers designed should be specific to the silenced gene and outside the gene fragment that was cloned into the pCFV plasmid. The sizes of amplified fragments are typically about 200–300 bp.

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